Signals and Systems

Lecture1: Introduction to Signals and Systems

Instructor: Prof. Ting Zhang
Zhejiang University

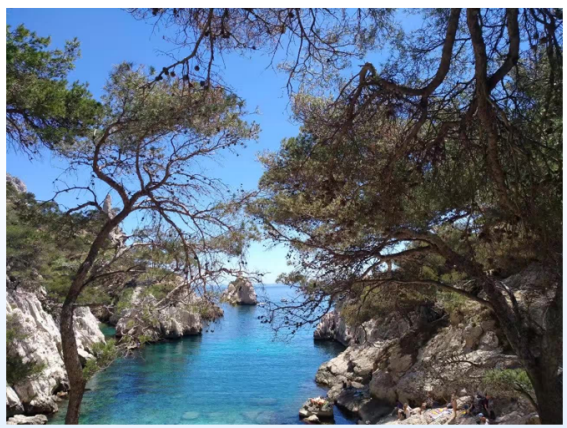
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Partly adapted from the materials provided on
the MIT OpenCourseWare

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2024~2025学年春夏学期

周二　第1~2节　紫金港校区西2-310

周四　第3~5节　紫金港校区教2-310

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Outline

1. Course Introduction
2. Introduction to Signals and Systems
3. Introduction to Signals
4. Introduction to Systems
5. Assignments

Course Introduction

This subject deals with mathematical methods used to describe signals and to analyze and synthesize systems

- Signals are variables that carry information
- Systems process input signals to produce output signals

Course Introduction

Why is it important?

Used almost everywhere

- ISEE: communications, circuit design, video/image, sonar/radar, speech processing
- Commercial electronics
- Aeronautics and astronautics
- Seismology
- Biomedical engineering
- Energy generation and distribution
- Chemical process control
- Financial analysis

Course Introduction

What is it all about?

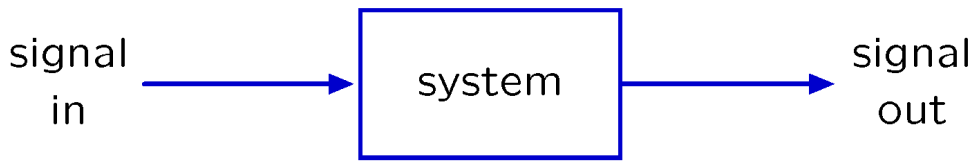
- Existing system analysis
- System design
  - Signal enhancement and restoration
  - Information extraction
  - Signal design
- System control

Outline

1. Course Introduction
2. Introduction to Signals and Systems
   - Example: Mass and Spring
   - Example: Tanks
   - Example: Cell Phone System
   - Signals and Systems
3. Introduction to Signals
4. Introduction to Systems
5. Assignments

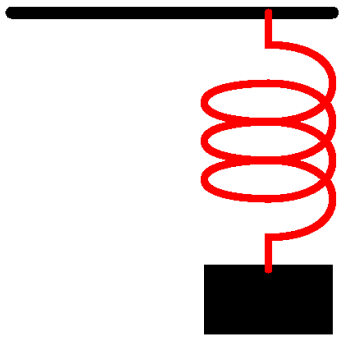
The Signals and Systems Abstraction

Describe a **system** (physical, mathematical, or computational) by the way it transforms an **input signal** into an **output signal**.

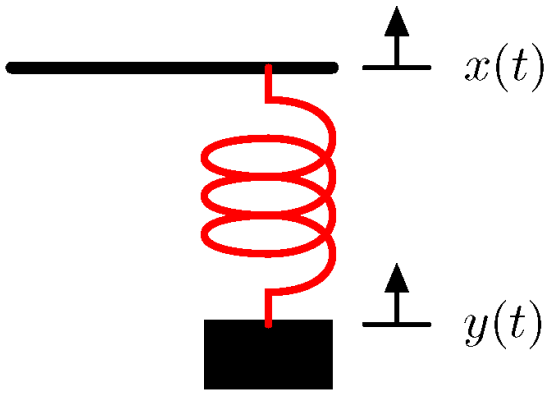


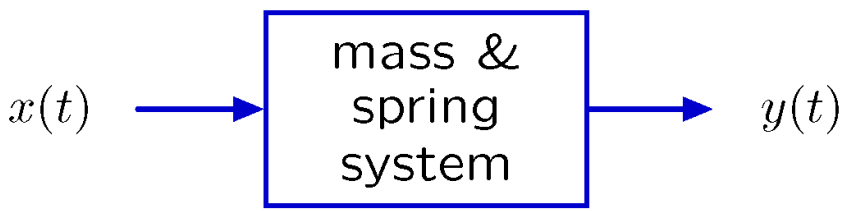
- **Physical**: Exists in the real world, governed by physical laws.
- **Mathematical**: Described by equations, state-space models, and transformations.
- **Computational**: Implemented via algorithms, digital processing, and machine learning.

Example: Mass and Spring

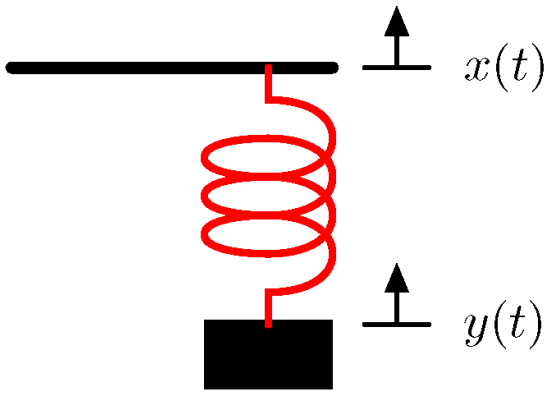


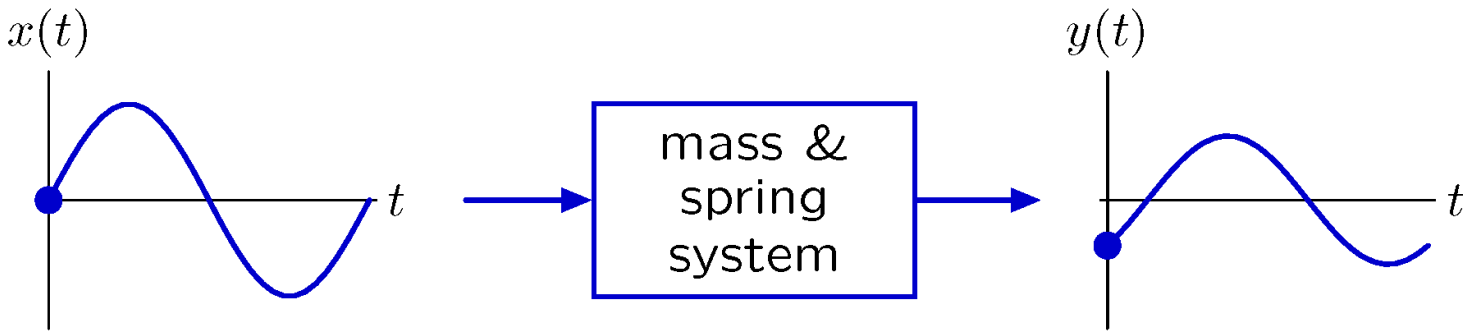
Example: Mass and Spring



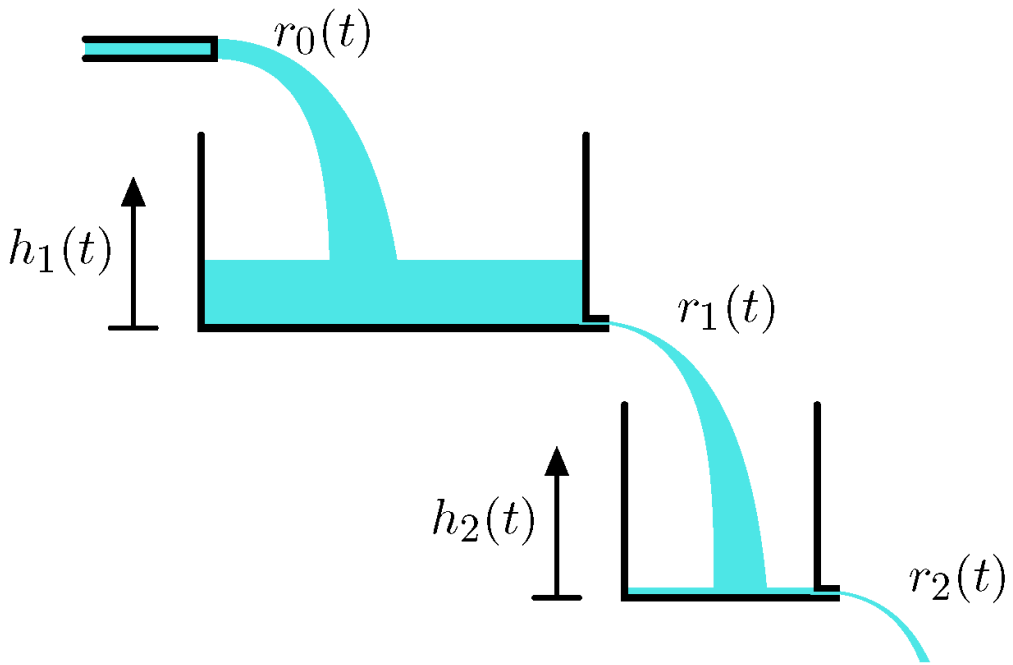


Example: Mass and Spring



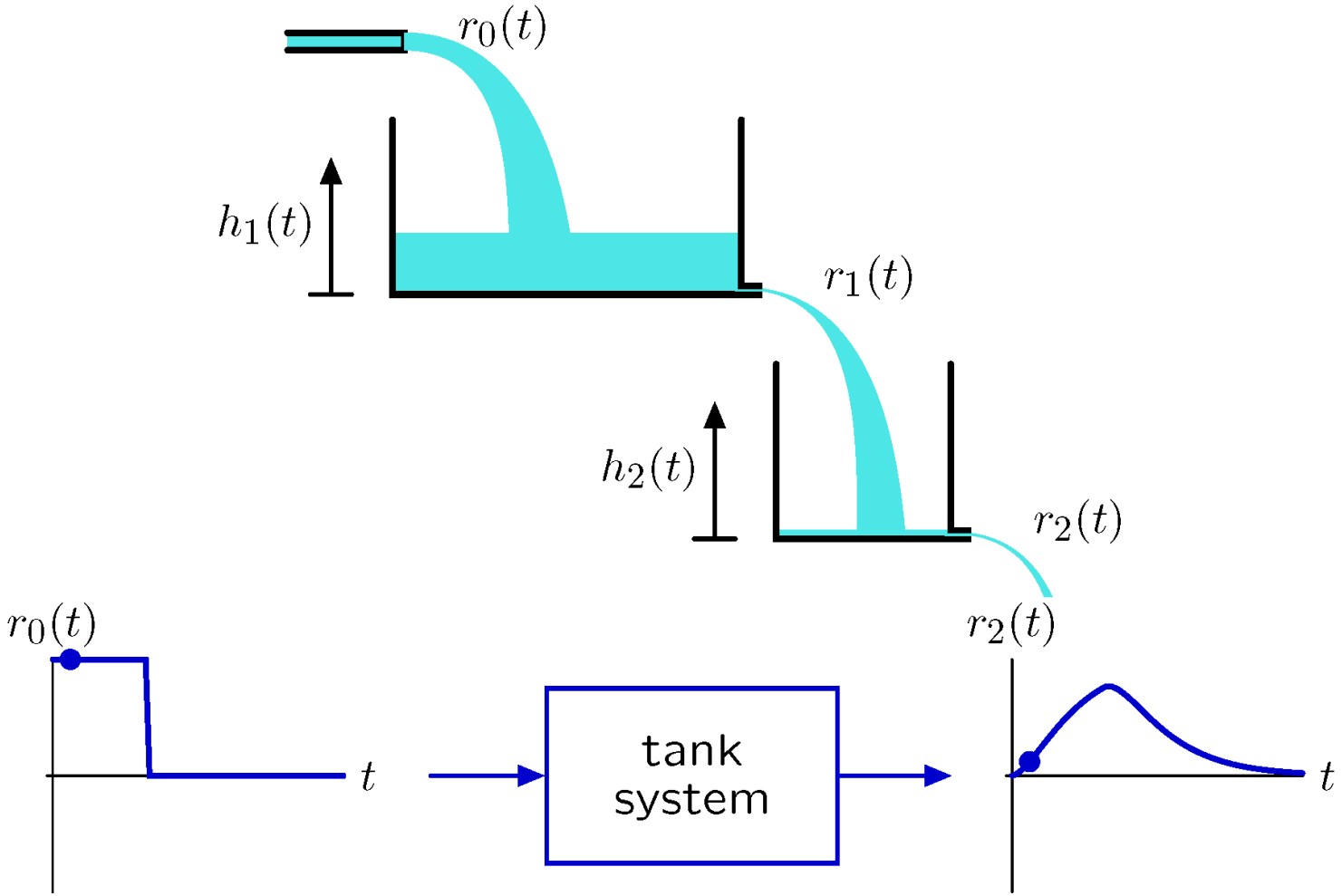


Example: Tanks

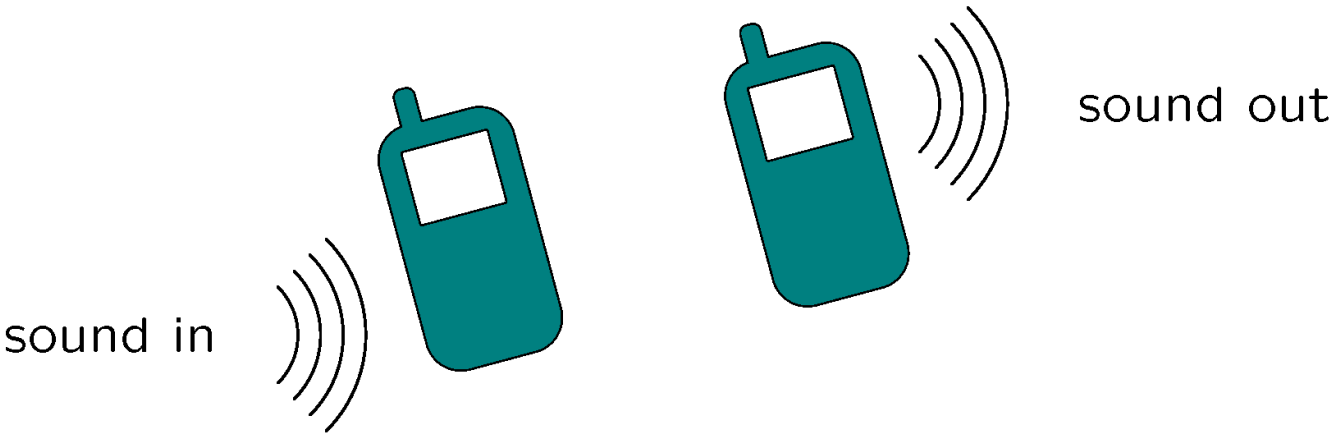


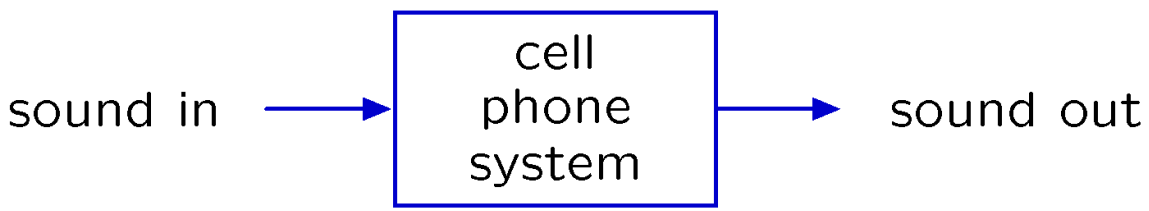


Example: Tanks

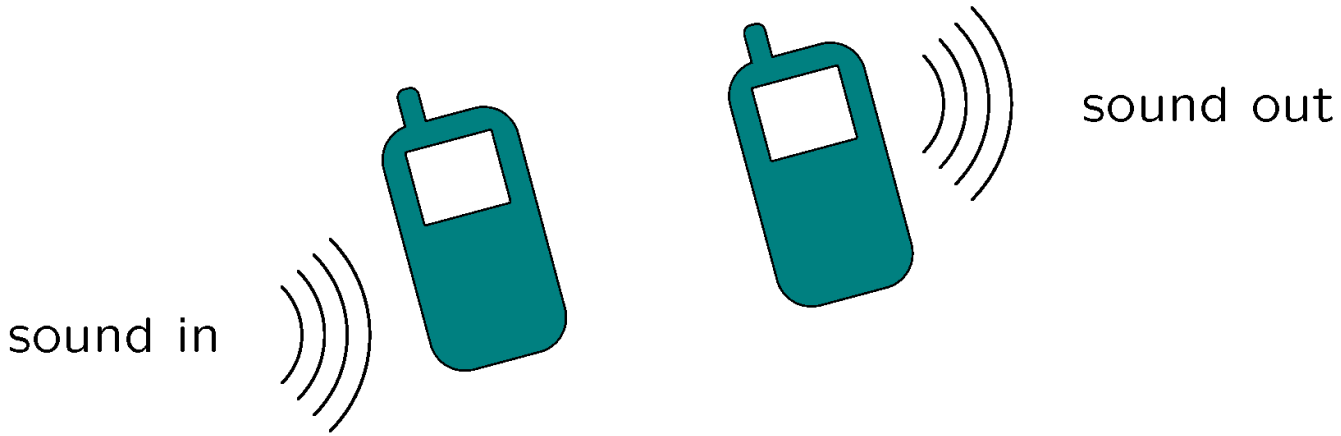


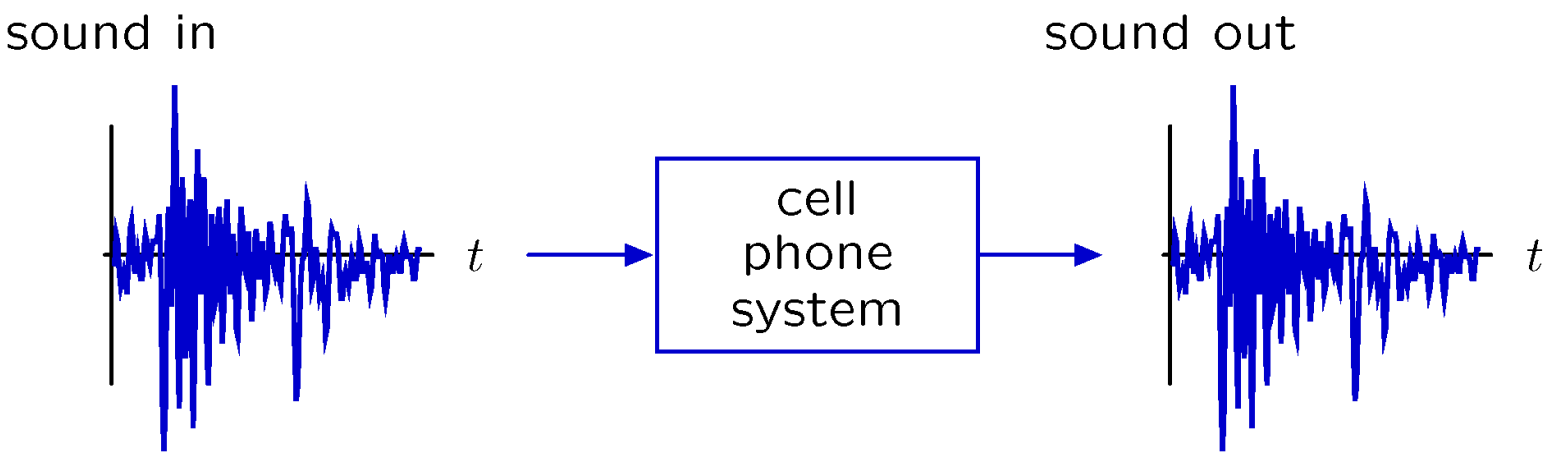
Example: Cell Phone System





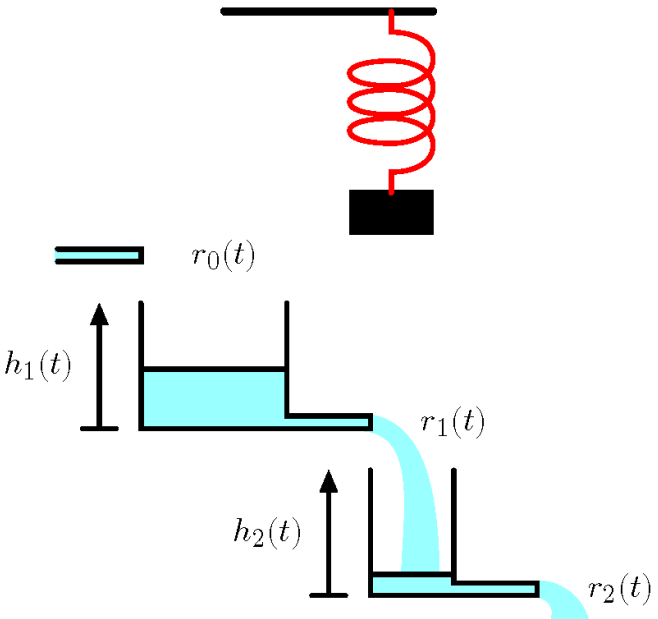
Example: Cell Phone System

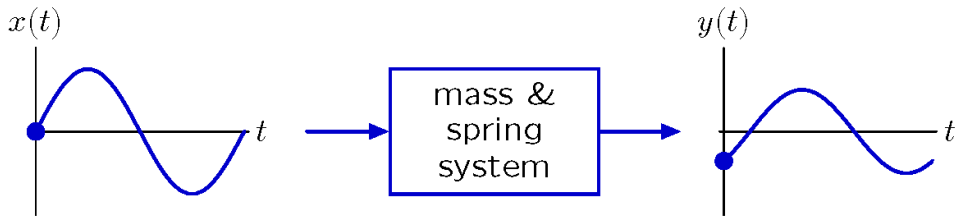


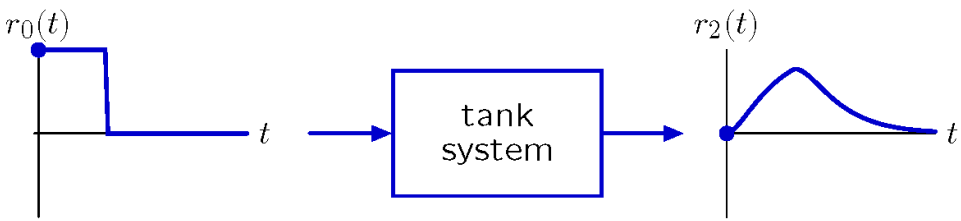


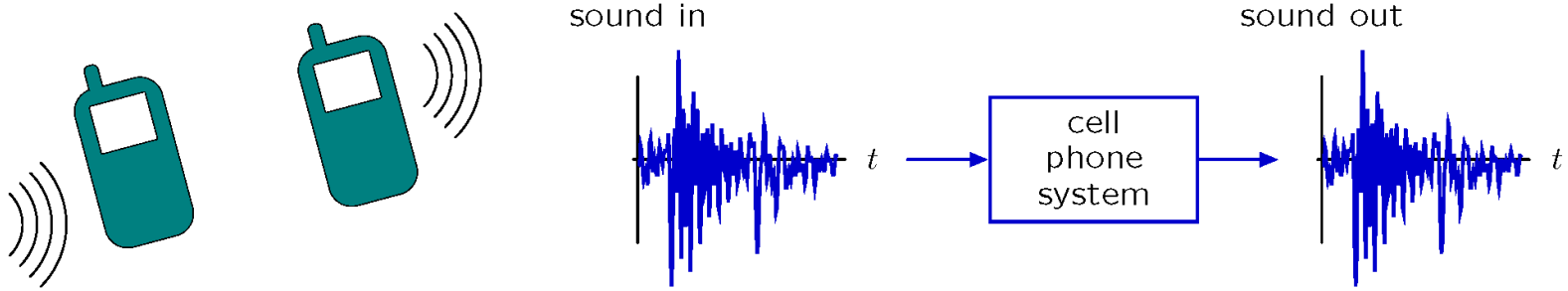
Signals and Systems: Widely Applicable

The Signals and Systems approach has broad application: electrical, mechanical, optical, acoustic, biological, financial, ...



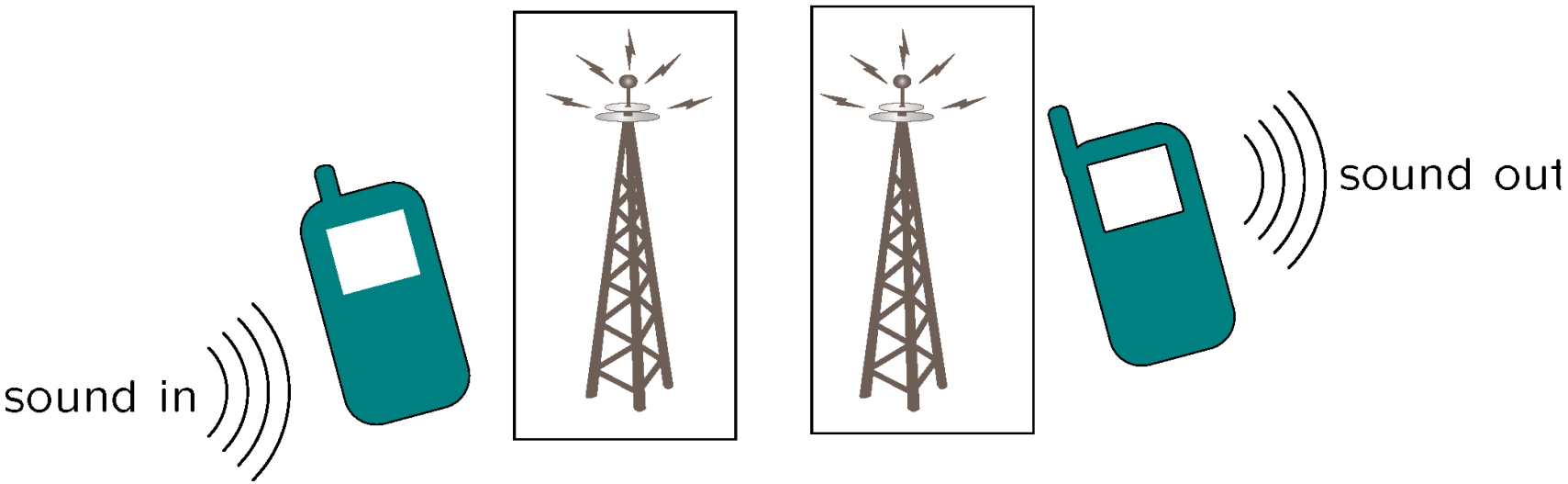


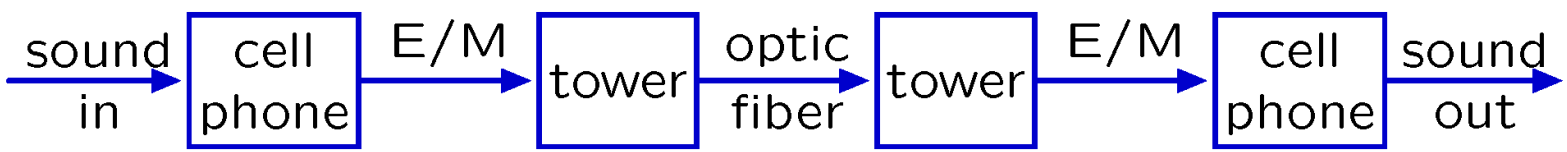




Signals and Systems: Modular

The representation does not depend upon the physical substrate.



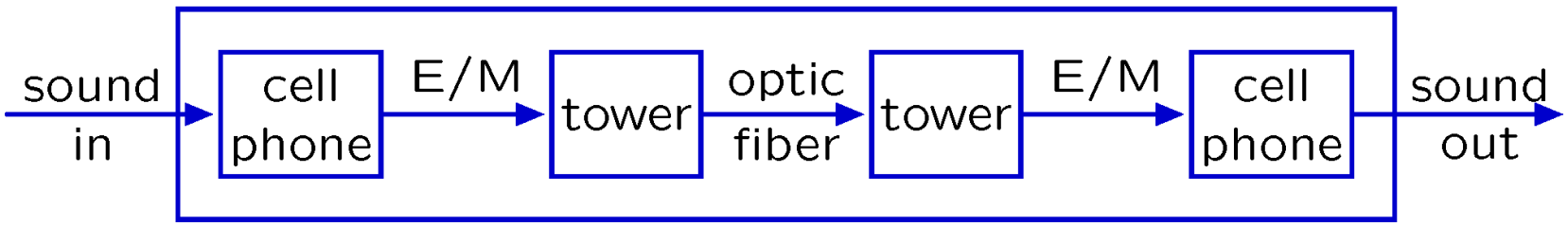


focuses on the flow of **information**, abstracts away everything else

Signals and Systems: Hierarchical

Representations of component systems are easily combined.

Example: cascade of component systems



Composite system



Component and composite systems have the same form, and are analyzed with same methods.

Neural Networks as a System

A neural network (NN) is a system that:

- Maps inputs to outputs using a network of interconnected neurons.
- Learns complex patterns through training.
- Can be represented as a computational, mathematical, or physical system.

NN as a Computational System

Definition: A neural network can be viewed as a digital system that processes information numerically.
Characteristics:

- Runs on CPUs, GPUs, or TPUs.
- Uses optimization techniques (e.g., gradient descent).
- Examples: Image classification, speech recognition, NLP.

NN as a Mathematical System

Definition: Neural networks are functions that approximate relationships between input and output.
Equation:

$$y = f(x, \theta)$$

where x is the input, θ represents weights, and f is the activation function.
Example: A simple feedforward neural network:

$$y = \sigma(Wx + b)$$

NN as a Physical System

Neural Networks for Wave Propagation

Wave Equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

- Describes acoustic, seismic, and electromagnetic wave propagation.
- Traditional solvers use numerical methods (FDM, FEM).
- Neural networks can approximate solutions using physics-informed learning.

Applications:

- **Acoustics**: Predicting sound field propagation.
- **Seismic Waves**: Modeling underground wave propagation.
- **Electromagnetic Waves**: Learning Maxwell's equations.

NN as a Composite System

A neural network consists of multiple components:

- **Neuron**: Basic unit of computation.
- **Layers**: Group of neurons performing transformations.
- **Network**: Interconnected layers forming a composite system.

Image generated by GPT[1]

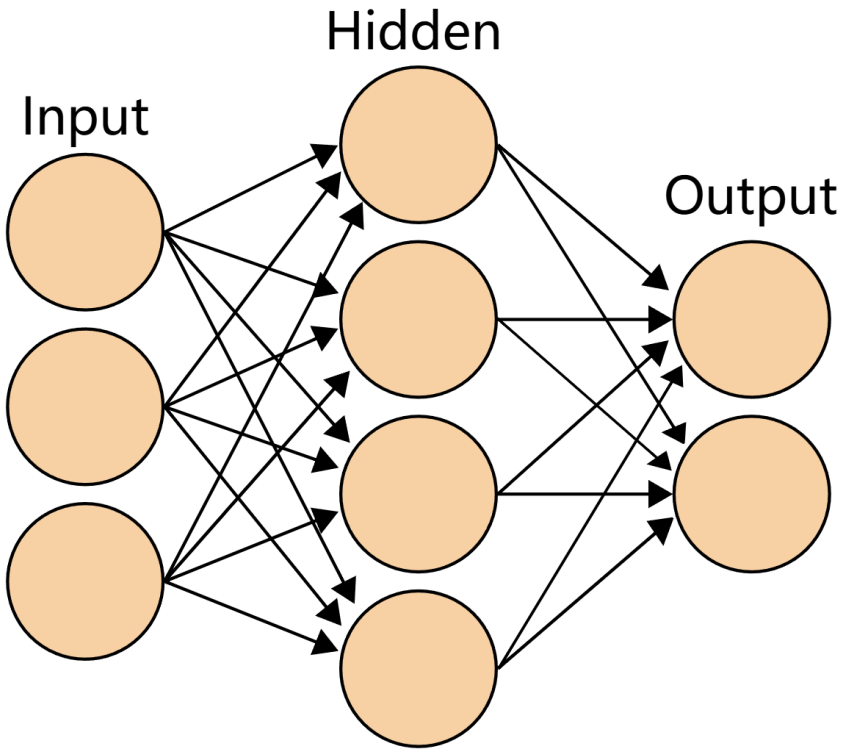


Figure: A typical deep neural network architecture.

Conclusion for NN

- Neural networks are versatile systems; They can be viewed as computational, mathematical, and physical entities.
- **Future Trends: AI for Science**
  - **AI-driven Scientific Discovery**: Accelerating research in physics, chemistry, and biology through deep learning models.
  - **Physics-Informed Neural Networks (PINN)**: Solving differential equations and simulating complex systems with constraints from physical laws.
  - **AI-enhanced Experimental Design**: Optimizing experiments in material science, drug discovery, and quantum computing.
  - **Scientific Generative Models**: AI-generated hypotheses, molecular structures, and protein folding predictions (e.g., AlphaFold).
  - **Neuromorphic and Quantum AI**: Advancing computation for high-dimensional scientific problems and energy-efficient AI models.

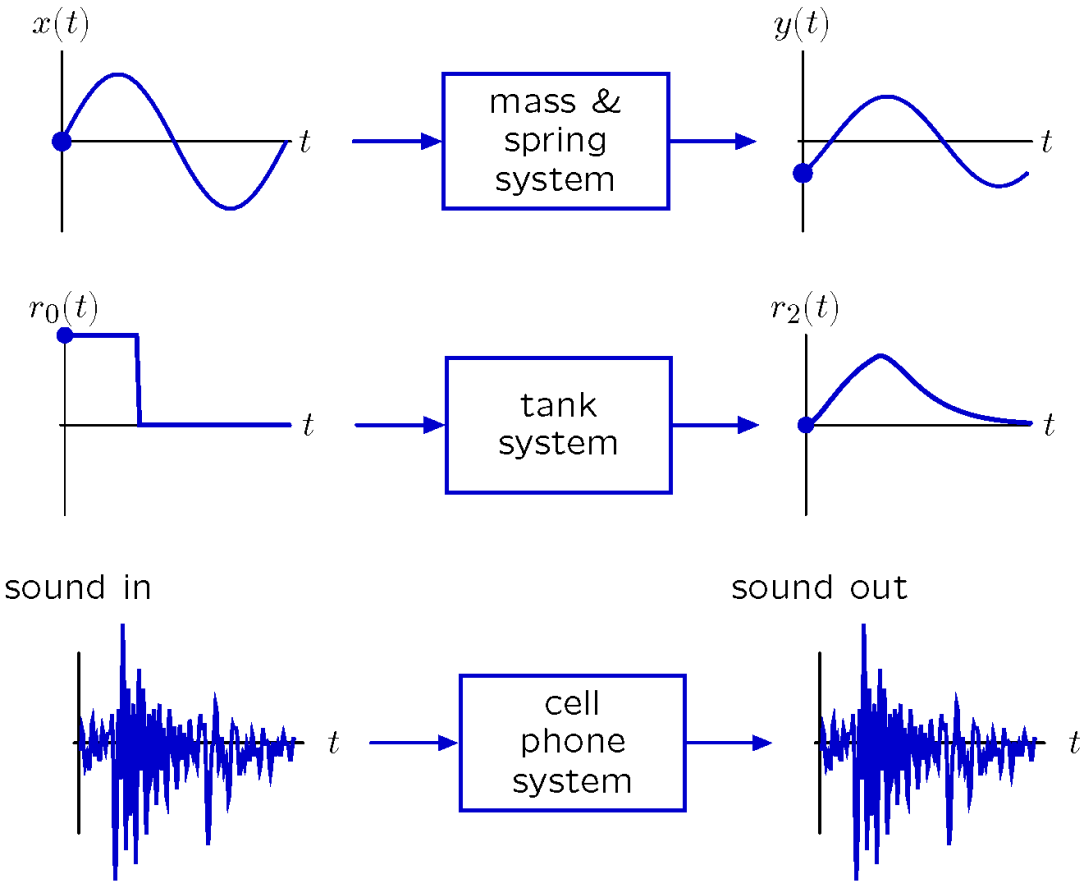
Outline

1. Course Introduction
2. Introduction to Signals and Systems
3. Introduction to Signals
   - Classification
   - Building-Block Signals
   - Transformations of time
4. Introduction to Systems
5. Assignments

Signals

Signals are mathematical functions.

- independent variable = time
- dependent variable = voltage, flow rate, sound pressure



Signals

Generic time

The term time is often used generically to represent the independent variable of a signal. The independent variable may be

- continuous or discrete
- 1-D, 2-D, $\cdots$ N-D

For this course: Focus on a single (1-D) independent variable which we call time.

What is a Dependent Variable?

- In machine learning, the **dependent variable** (also called the target variable) is what the model aims to predict.
- It depends on one or more **independent variables** (features).
- Examples:
  - Regression: Predicting house price ($) based on size, location, etc.
  - Classification: Predicting spam (1) or not spam (0) based on email content.
  - Time Series: Forecasting stock prices based on past trends.

Example 1: Regression - House Price Prediction

Problem: Predicting house prices based on various factors.

- **Dependent Variable:** House Price (Y)
- **Independent Variables:** Square footage, number of bedrooms, location, etc.

Model: Linear Regression

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + ... + \varepsilon$$

Example 2: Classification - Email Spam Detection

Problem: Classifying emails as spam or not spam.

- **Dependent Variable:** Spam (1) or Not Spam (0)
- **Independent Variables:** Frequency of certain words, presence of links, sender reputation, etc.

Model: Logistic Regression / Decision Trees / Neural Networks

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots)}}$$

Example 3: Time Series - Stock Price Prediction

Problem: Predicting future stock prices based on historical data.

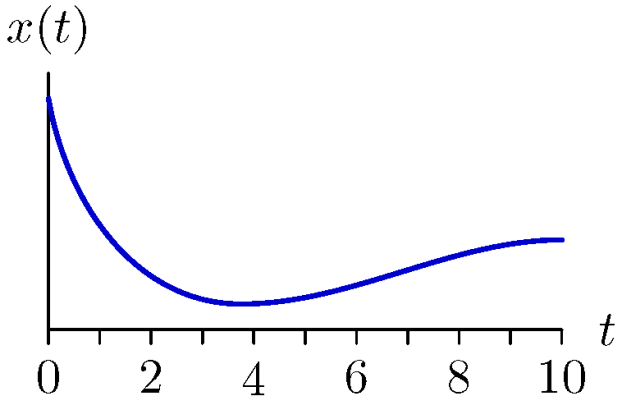
- **Dependent Variable:** Future Stock Price (Y_t)
- **Independent Variables:** Past stock prices, trading volume, news sentiment.

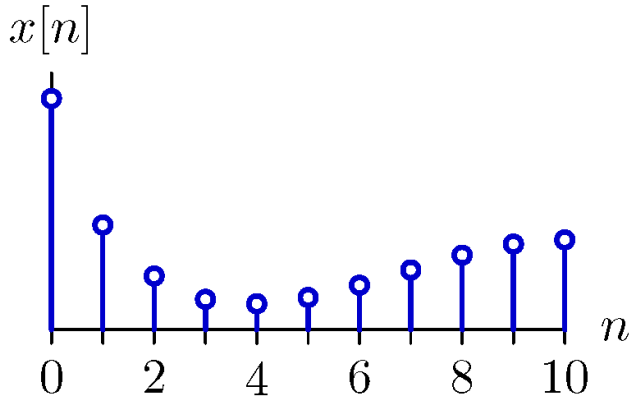
Model: Recurrent Neural Networks (RNN), ARIMA

$$Y_t = f(Y_{t-1}, Y_{t-2}, \ldots, X_t)$$

Signals: Continuous Time and Discrete Time

continuous "time" (CT) and discrete "time" (DT)





Many physical systems operate in continuous time.

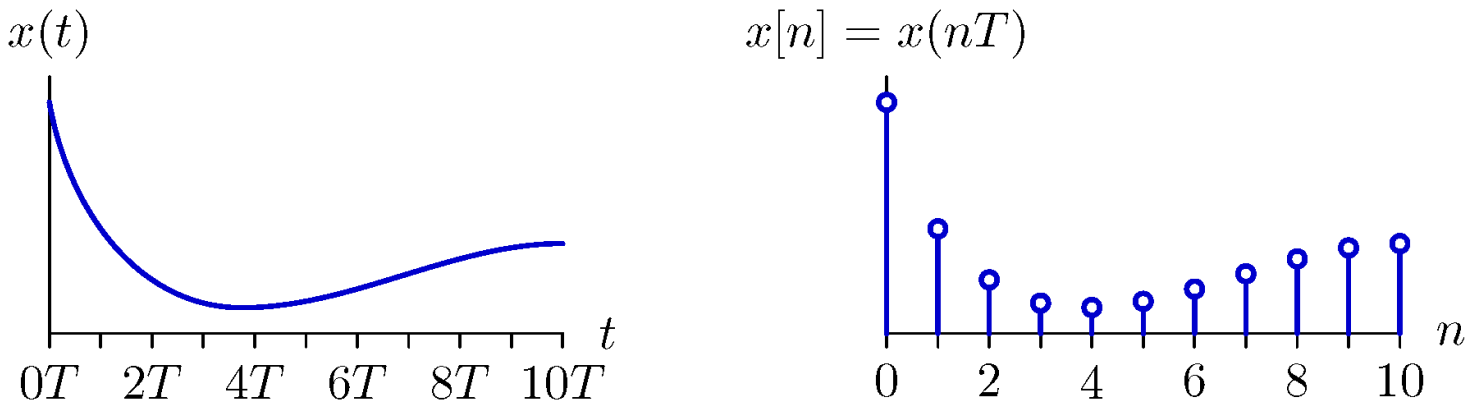
- mass and spring
- leaky tank

Digital computations are done in discrete time.

- state machines: given the current input and current state, what is the next output and next state.

Signals: Continuous Time and Discrete Time

Sampling: converting CT signals to DT



$T =$ sampling interval

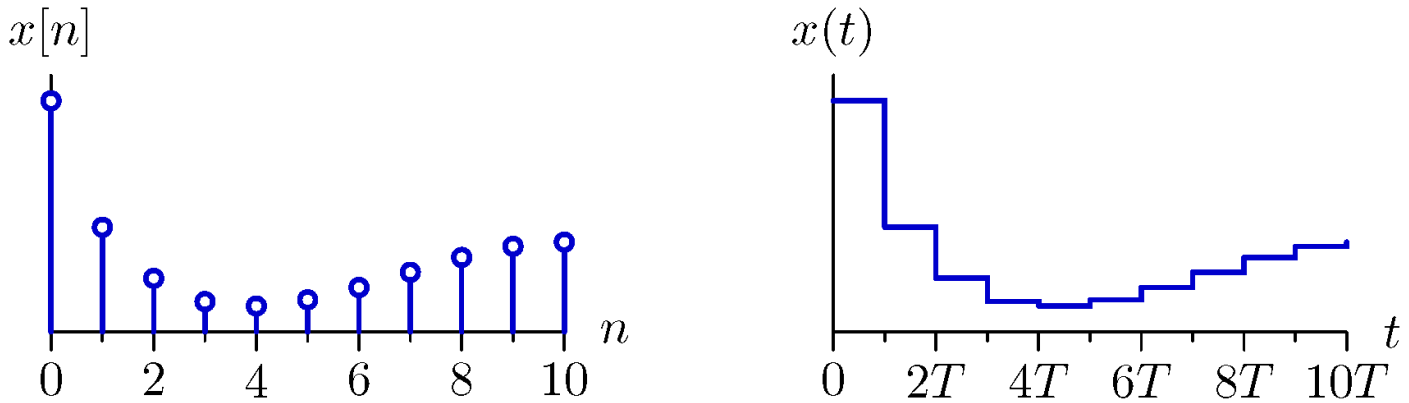
Important for computational manipulation of physical data.

- digital representations of audio signals (e.g., MP3)
- digital representations of pictures (e.g., JPEG)

Signals: Continuous Time and Discrete Time

Reconstruction: converting DT signals to CT

zero-order hold



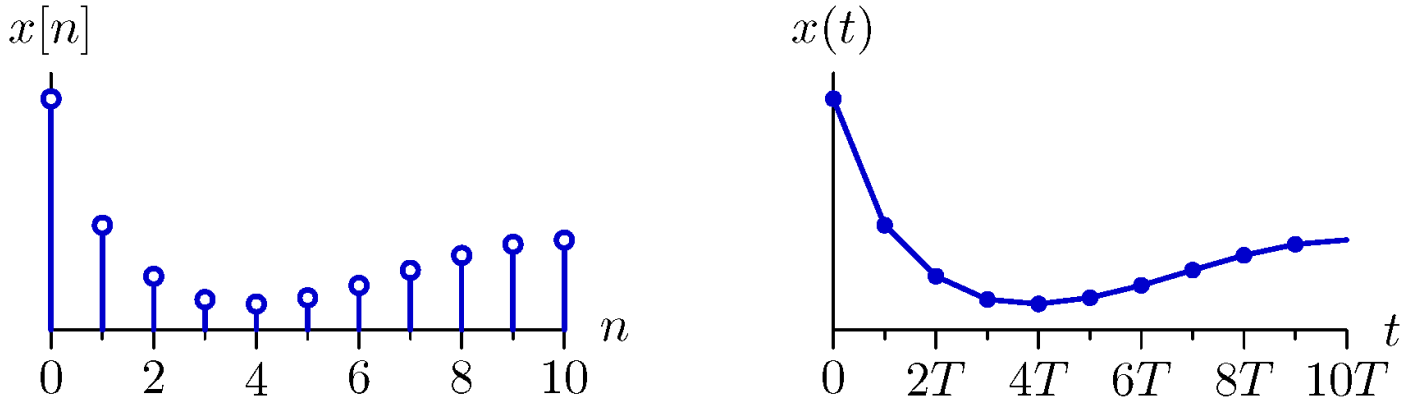
$T =$ sampling interval

commonly used in audio output devices such as CD players

Signals: Continuous Time and Discrete Time

Reconstruction: converting DT signals to CT

piecewise linear



$T =$ sampling interval

commonly used in rendering images

Signals: Real and Complex

Signals can be real, imaginary, or complex. An important class of signals are the complex exponentials:

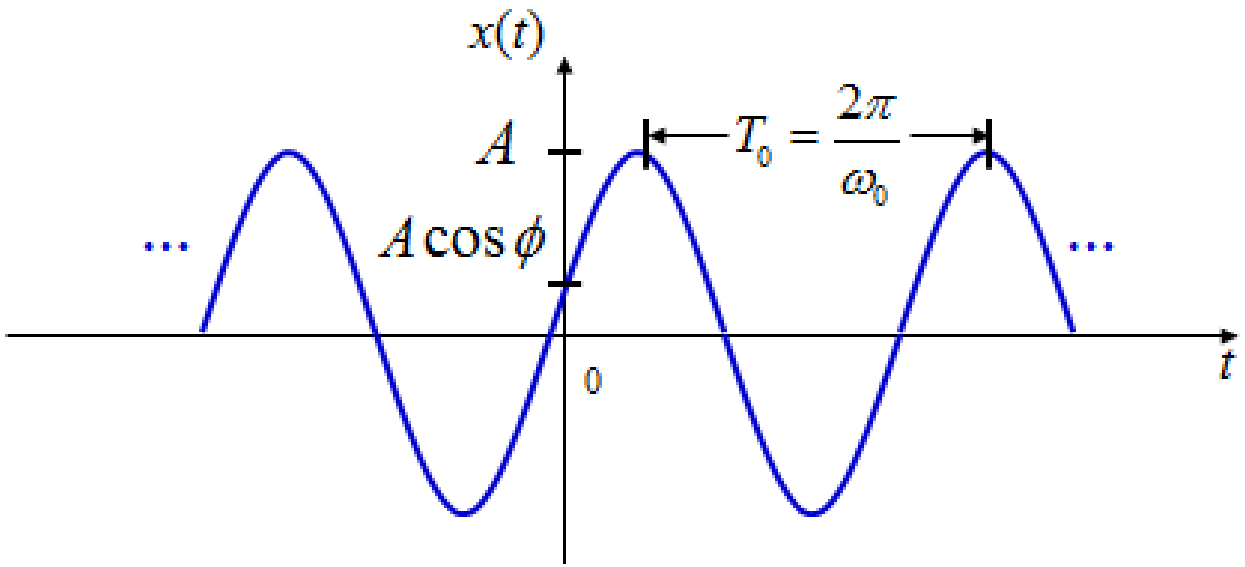
- the CT signal $x(t) = e^{st}$ where s is a complex number,
- the DT signal $x[n] = z^n$ where z is a complex number.

Q. Why do we deal with complex signals?

A. They are often analytically simpler to deal with than real signals.

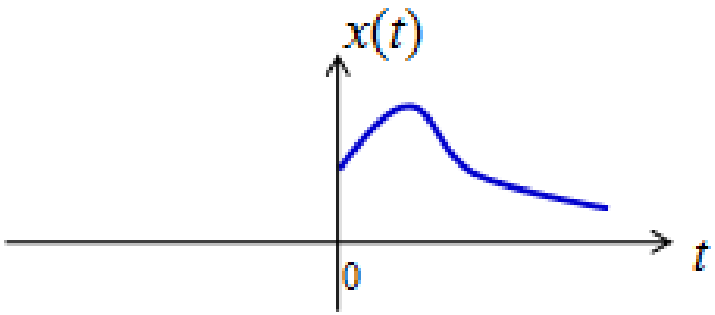
Signals: Periodic and Aperiodic

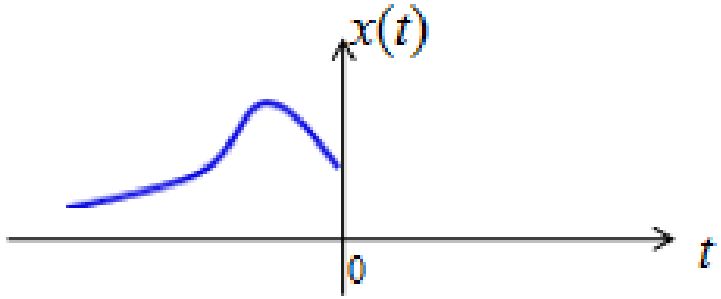
Periodic signals have the property that $x(t+T) = x(t)$ for all t. The smallest value of T that satisfies the definition is called the period.



Signals: Causal and Anti-causal

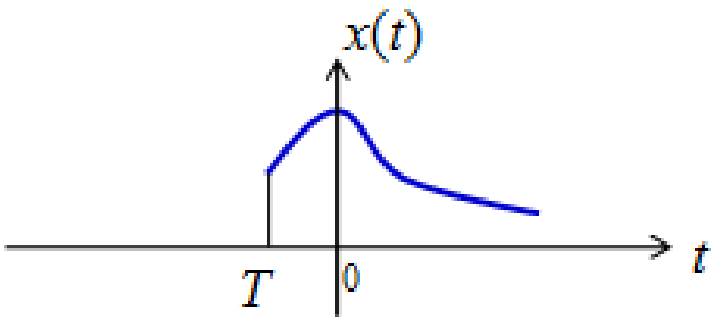
A causal signal is zero for $t < 0$ and anti-causal signal is zero for $t > 0$

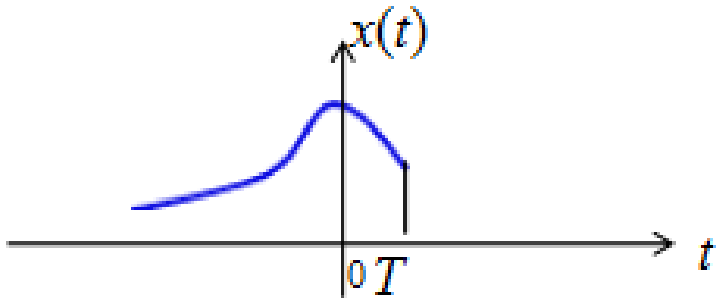




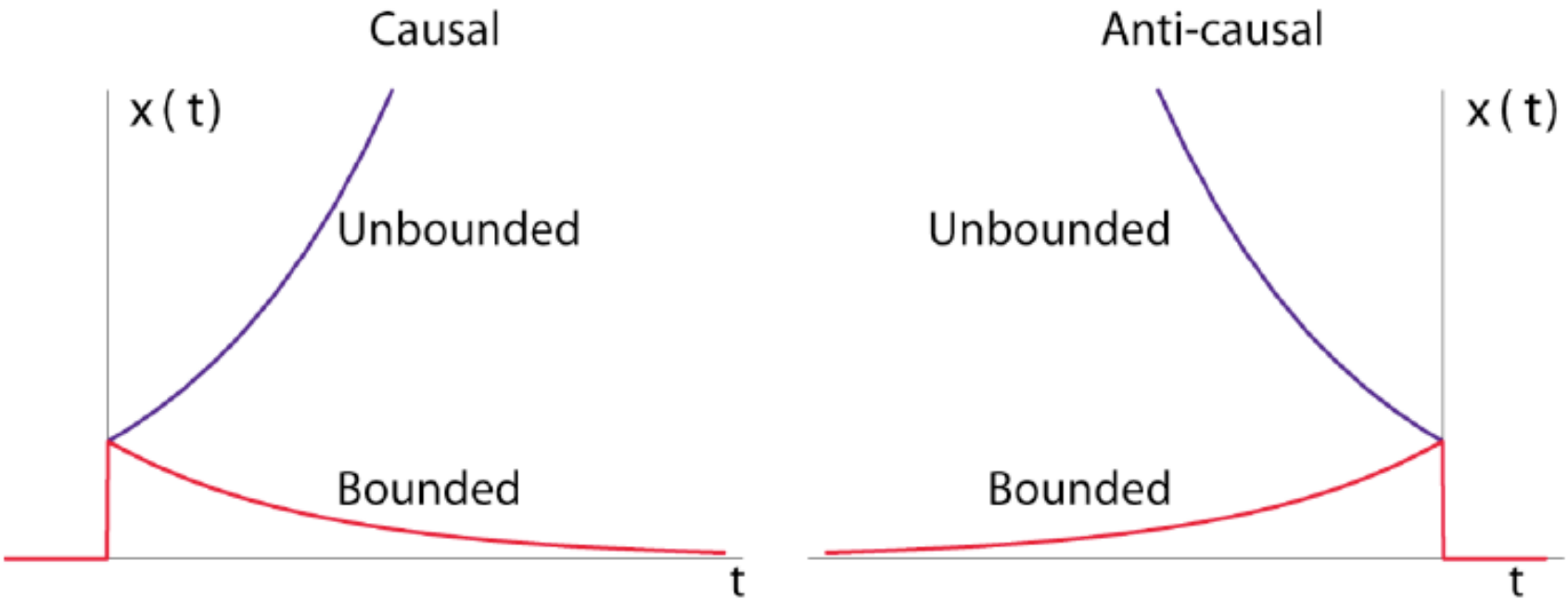
Signals: Right- and Left-sided

A right-sided signal is zero for $t < T$ and left-sided signal is zero for $t > T$ where T can be positive or negative.





Signals: Bounded and Unbounded



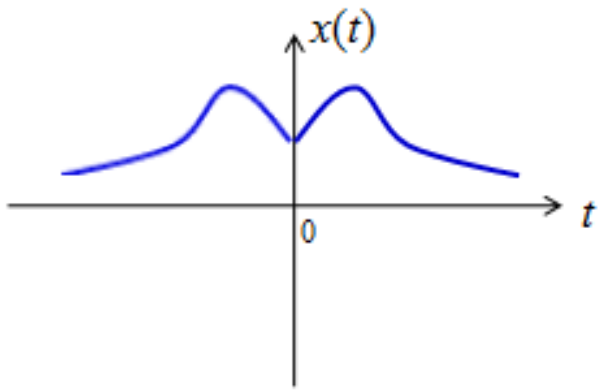
Whether the output signal of a system is bounded or unbounded determines the stability of the system.

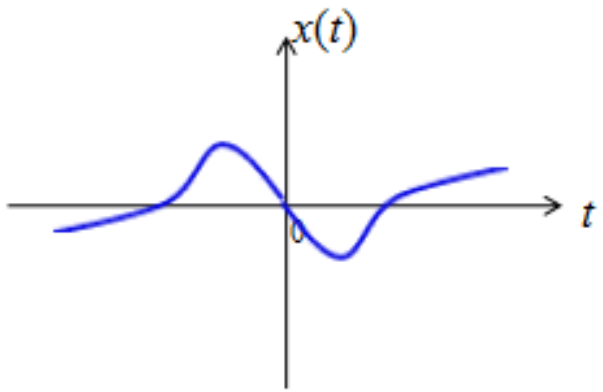
Signals: Even and Odd

$$x_e(t) = x_e(-t) \qquad x_o(t) = -x_o(-t)$$

- Any signal can be broken into a sum of even and odd signals

$$Ev\{x(t)\} = \frac{x(t) + x(-t)}{2} \qquad Od\{x(t)\} = \frac{x(t) - x(-t)}{2}$$





Signals: Energy and Power

- Energy and Power Signals
  - CT signals

$$E_\infty = \lim_{T\to\infty}\int_{-T}^{T} |\, x(t)\,|^2 dt = \int_{-\infty}^{+\infty} |\, x(t)\,|^2 dt$$

$$P_\infty = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T} |\, x(t)\,|^2 dt$$

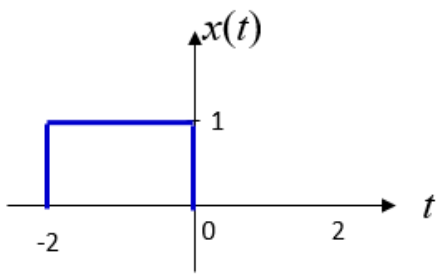
  - DT signals

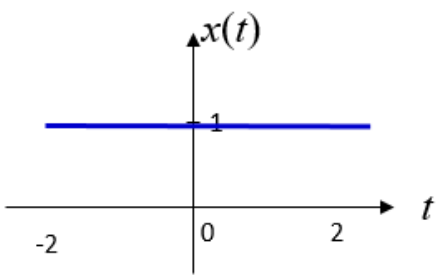
$$E_\infty = \lim_{N\to\infty}\sum_{n=-N}^{+N} |\, x[n]\,|^2 = \sum_{n=-\infty}^{+\infty} |\, x[n]\,|^2$$

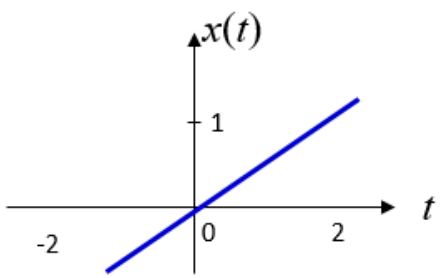
$$P_\infty = \lim_{N\to\infty}\frac{1}{2N+1}\sum_{n=-N}^{+N} |\, x[n]\,|^2$$

Signals: Energy and Power

- Energy and Power Signals
  - Finite total energy $E_\infty < \infty \Rightarrow P_\infty = \lim_{T\to\infty} \frac{E_\infty}{2T} = 0$
  - Finite average power $P_\infty < \infty \Rightarrow E_\infty = \infty$
  - Neither E_∞ nor P_∞ are finite



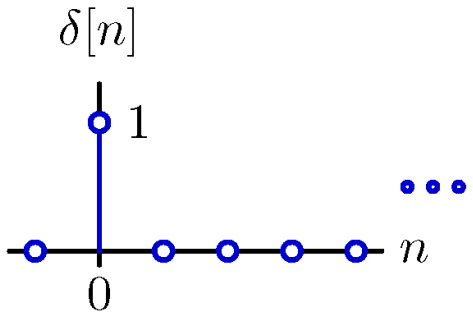




Elementary Building-Block Signals

Elementary DT signal: $\delta[n]$.

$$\delta[n] = \begin{cases} 1, & \text{if } n = 0; \\ 0, & \text{otherwise} \end{cases}$$



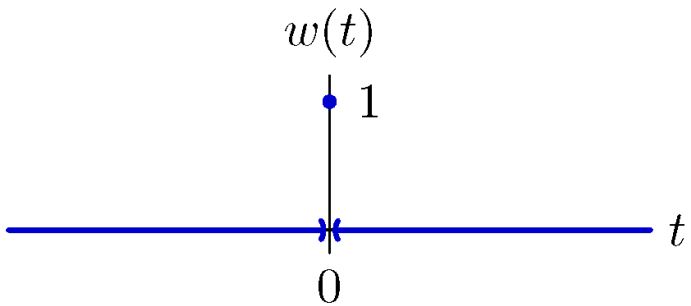
- shortest possible duration (most "transient")
- useful for constructing more complex signals

What CT signal serves the same purpose?

Elementary CT Building-Block Signal

Consider the analogous CT signal.

$$w(t) = \begin{cases} 0 & t < 0 \\ 1 & t = 0 \\ 0 & t > 0 \end{cases}$$

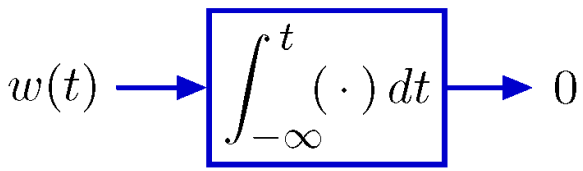


Is this a good choice as a building-block signal?

Elementary CT Building-Block Signal

Consider the analogous CT signal.

$$w(t) = \begin{cases} 0 & t < 0 \\ 1 & t = 0 \\ 0 & t > 0 \end{cases}$$



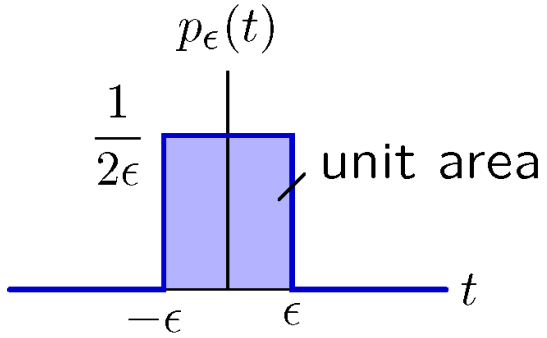
Is this a good choice as a building-block signal? No

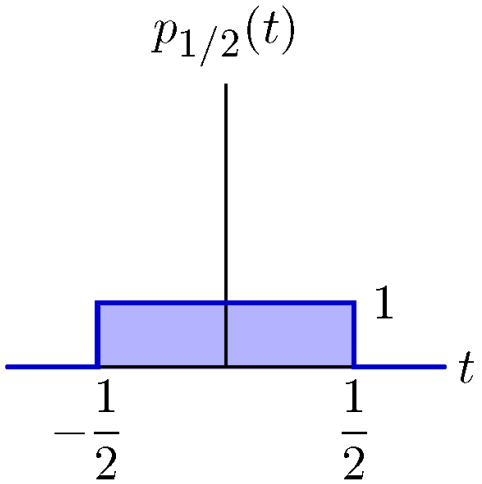
The integral of $w(t)$ is zero!

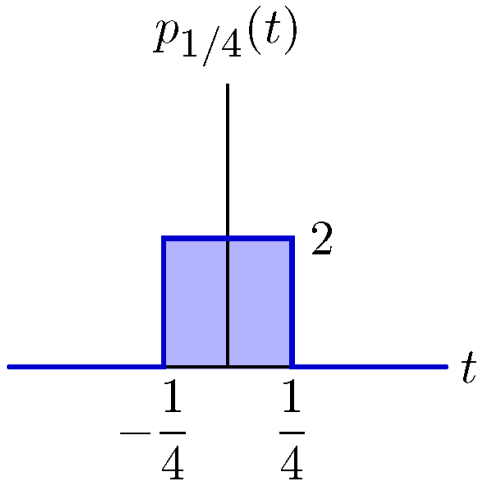
Unit-Impulse Signal

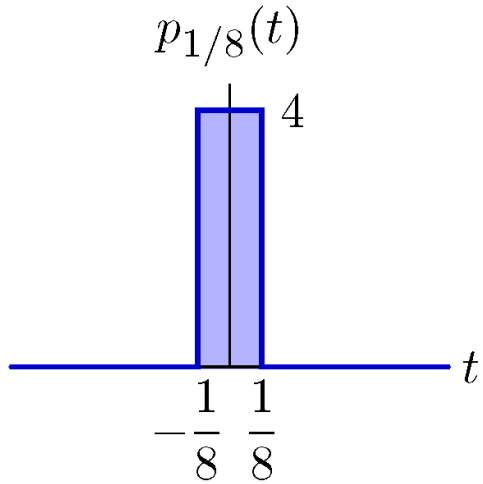
The unit-impulse signal acts as a pulse with unit area but zero width.

$$\delta(t) = \lim_{\epsilon \to 0} p_\epsilon(t)$$



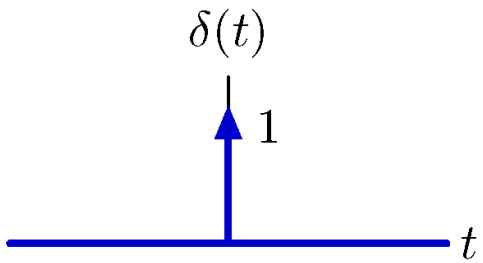






Unit-Impulse Signal

The unit-impulse function is represented by an arrow with the number **1**, which represents its area or "weight."

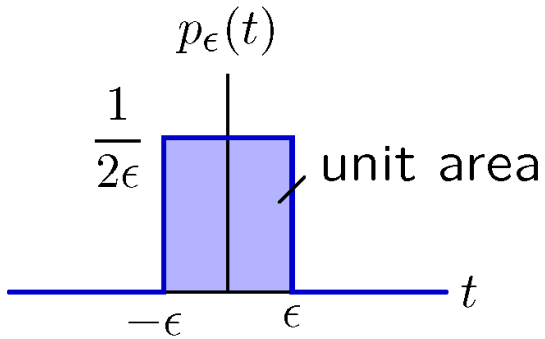


It has two seemingly contradictory properties:

- it is nonzero only at $t = 0$, and
- its definite integral $(-\infty, \infty)$ is one!

Both of these properties follow from thinking about $\delta(t)$ as a limit:

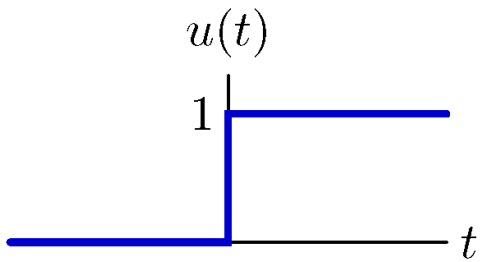
$$\delta(t) = \lim_{\epsilon \to 0} p_\epsilon(t)$$



Unit-Impulse and Unit-Step Signals

The indefinite integral of the unit-impulse is the unit-step.

$$u(t) = \int_{-\infty}^{t} \delta(\lambda)\, d\lambda = \begin{cases} 1; & t \geq 0 \\ 0; & \text{otherwise} \end{cases}$$



Equivalently

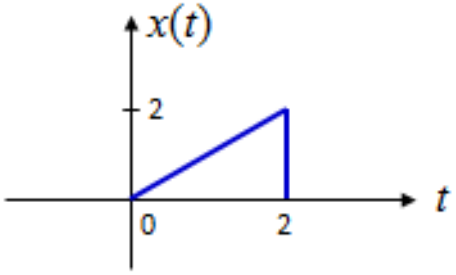


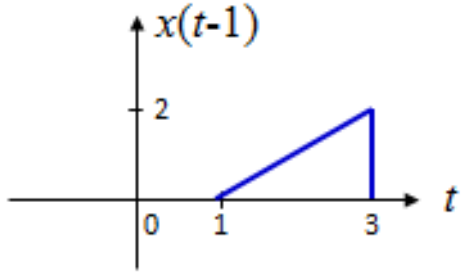
Transformations of Time

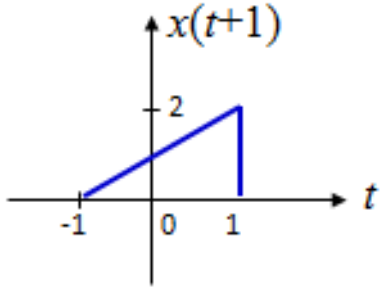
- Time shift

$x(t) \Rightarrow x(t-t_0)$ $t_0 > 0$ delayed, move to right

$t_0 < 0$ advanced, move to left

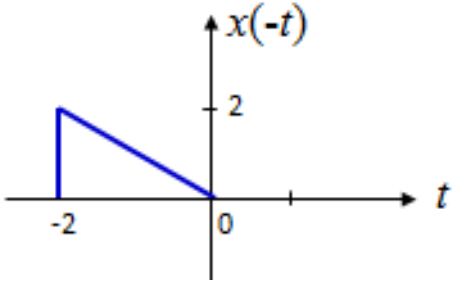


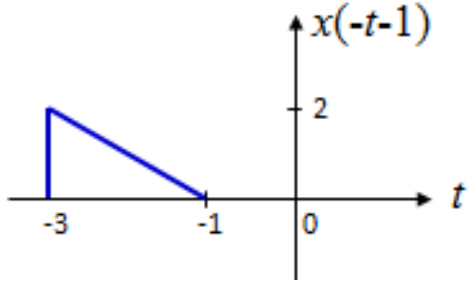


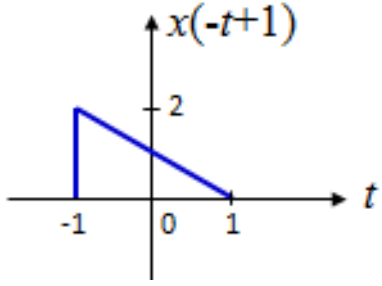


- Time reversal

$x(t) \Rightarrow x(-t)$



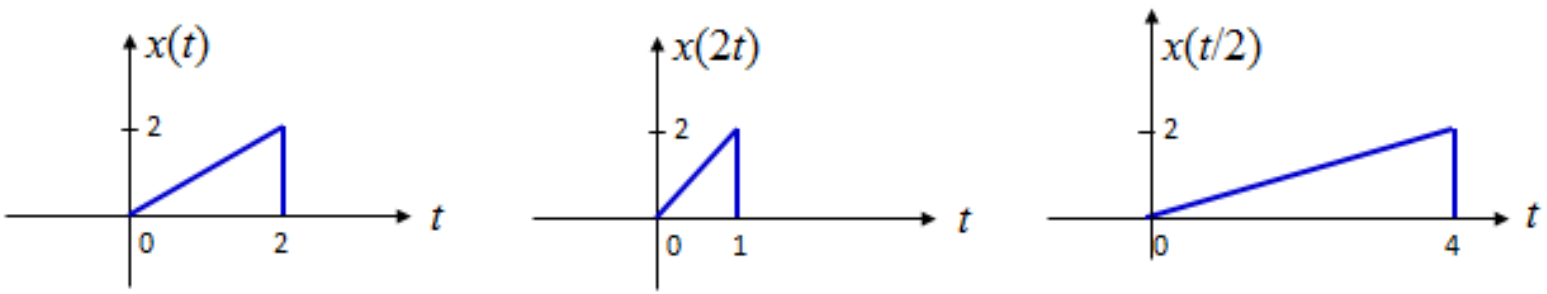




Transformations of time

- Time scaling

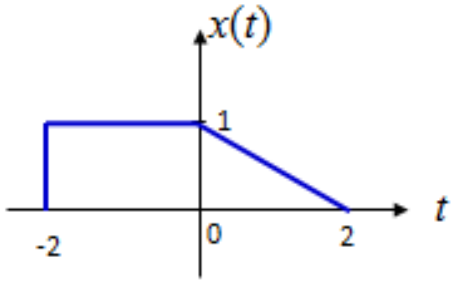
$$x(t) \Rightarrow x(\alpha t) \begin{cases} |\alpha| < 1 & \text{stretched} \\ |\alpha| > 1 & \text{compressed} \end{cases}$$



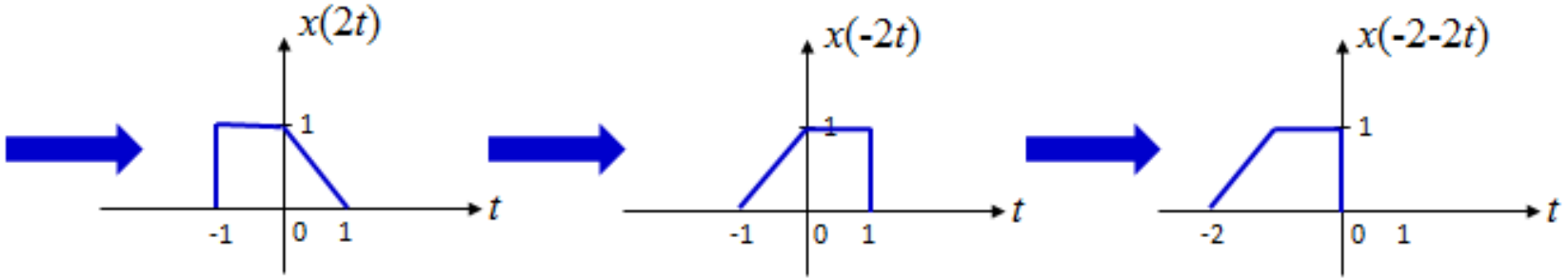
Time scaling of signals in DT?

Check Yourself

Given $x(t)$, sketch $x(-2-2t)$.



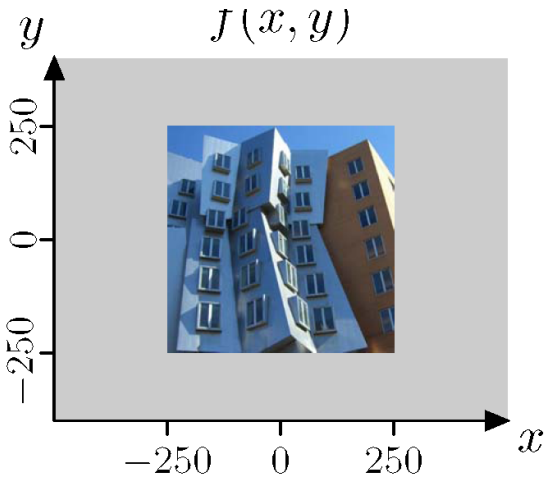
① $x(t) \Rightarrow x(2t) \Rightarrow x(-2t) \Rightarrow x(-2-2t)$



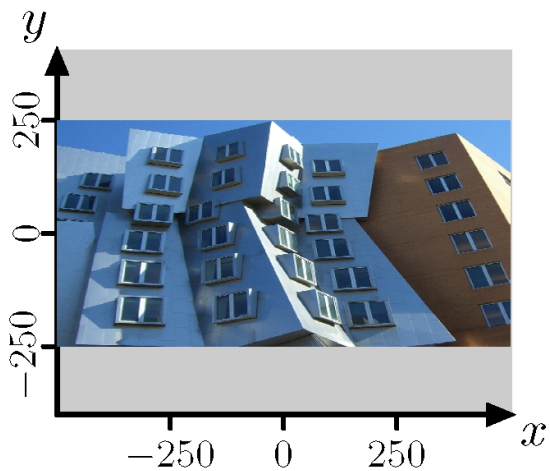
② $x(t) \Rightarrow x(2t) \Rightarrow x(2t-2) \Rightarrow x(-2t-2)$

③ ...

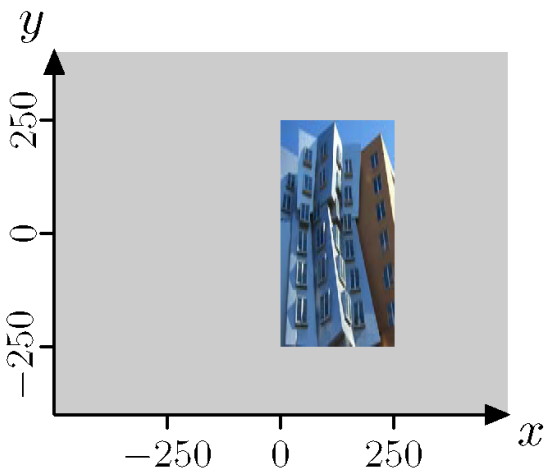
Check Yourself



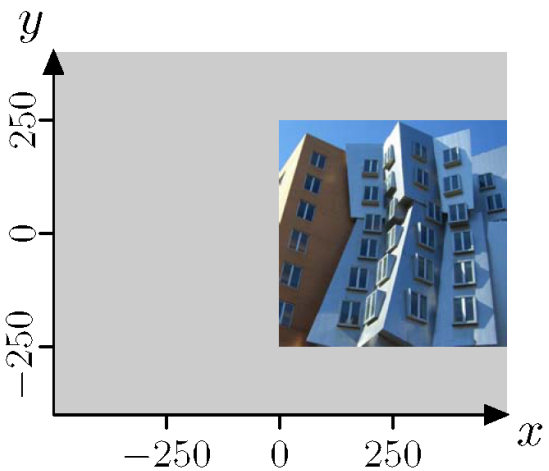
How many images match the expressions beneath them?



$f_1(x,y) = f(2x,y)$?

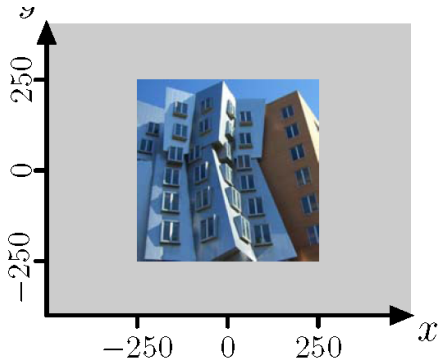


$f_2(x,y) = f(2x-250,y)$?

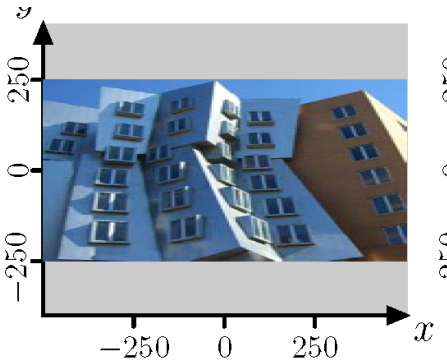


$f_3(x,y) = f(-x-250,y)$?

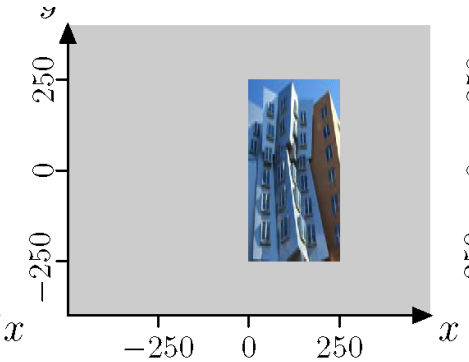
Check Yourself



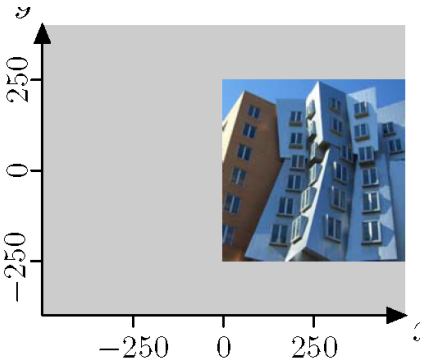
$f(x, y)$



$f_1(x, y) = f(2x, y)$?



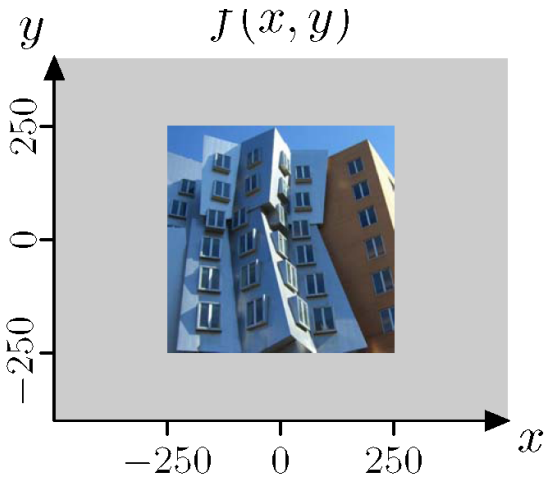
$f_2(x, y) = f(2x-250, y)$?



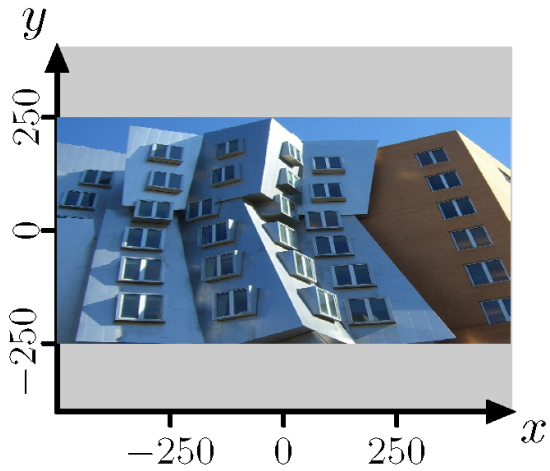
$f_3(x, y) = f(-x-250, y)$?

$$
\begin{aligned}
x = 0 \quad &\rightarrow f_1(0, y) = f(0, y) && \surd \\
x = 250 \quad &\rightarrow f_1(250, y) = f(500, y) && \times \\
\\
x = 0 \quad &\rightarrow f_2(0, y) = f(-250, y) && \surd \\
x = 250 \quad &\rightarrow f_2(250, y) = f(250, y) && \surd \\
\\
x = 0 \quad &\rightarrow f_3(0, y) = f(-250, y) && \times \\
x = 250 \quad &\rightarrow f_3(250, y) = f(-500, y) && \times
\end{aligned}
$$

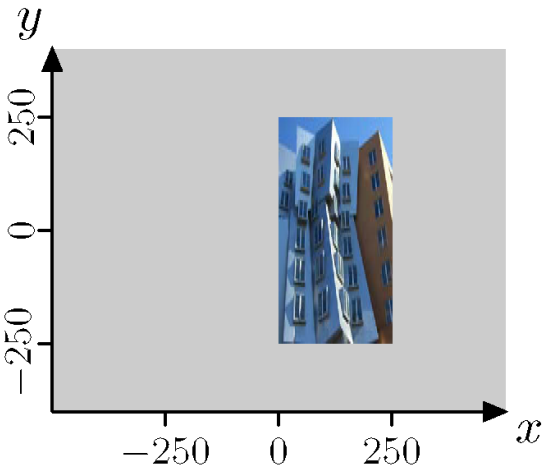
Check Yourself



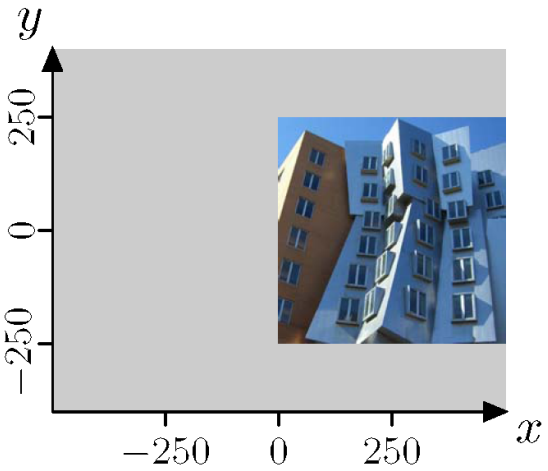
How many images match the expressions beneath them?



~~$f_1(x,y) = f(2x,y)$?~~



$f_2(x,y) = f(2x-250,y)$?



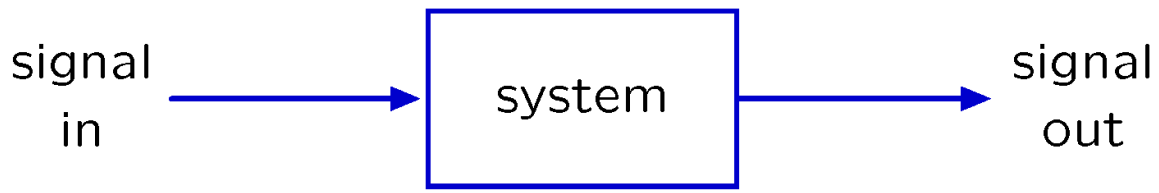
~~$f_3(x,y) = f($ $-250,y)$?~~

Outline

1. Course Introduction
2. Introduction to Signals and Systems
3. Introduction to Signals
4. Introduction to Systems
   - Classification
5. Assignments

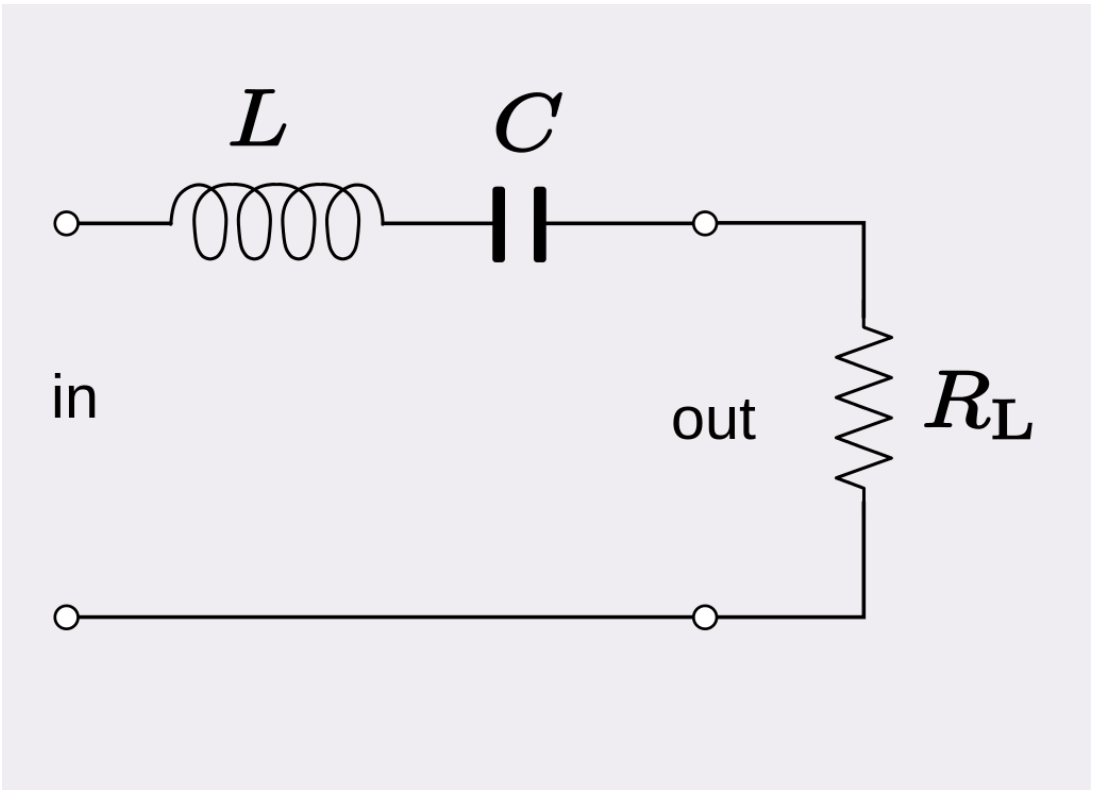
The Signals and Systems Abstraction

Describe a **system** (physical, mathematical, or computational) by the way it transforms an **input signal** into an **output signal**.



Introduction to RLC Circuits

- The RLC circuit consists of a Resistor (R), an Inductor (L), and a Capacitor (C) in series.
- It is widely used in signal processing, filters, and oscillators.
- The system can be modeled using a second-order differential equation.



Governing Differential Equation

Using Kirchhoff's Voltage Law (KVL):

$$V_{in}(t) = V_R + V_L + V_C \tag{1}$$

where:

- $V_R = Ri(t)$ (Ohm's Law)
- $V_L = L\frac{di}{dt}$ (Inductor Voltage)
- $V_C = \frac{1}{C}\int i(t)dt$ (Capacitor Voltage)

Taking the derivative,

$$\frac{d^2v_C}{dt^2} + \frac{R}{L}\frac{dv_C}{dt} + \frac{1}{LC}v_C = \frac{1}{L}\frac{dv_{in}}{dt} \tag{2}$$

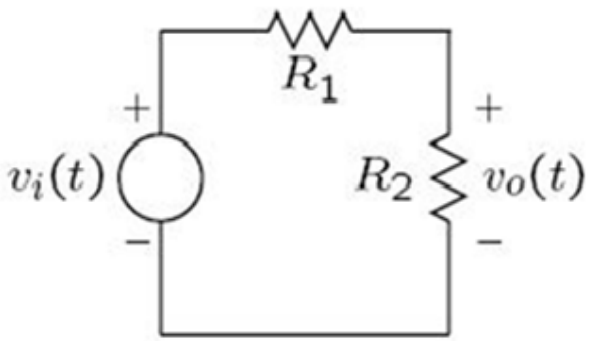
Systems: with and without memory

- **Memoryless sytem**

  The output of a memoryless sytem at a given time depends only on the input at the same time.

Ex.#1 $y[n] = (2x[n] - x^2[n])^2$

Ex.#2



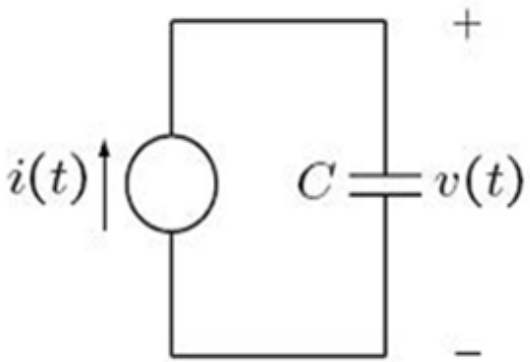
$$v_o(t) = \frac{R_2}{R_1 + R_2} v_i(t).$$

Systems: with and without memory

- **Systems with memory**

Ex.#1 $$y[n] = \sum_{k=-\infty}^{n} x[k]$$

Ex.#2



$$i(t) = C\frac{dv(t)}{dt},$$
$$v(t) = \frac{1}{C}\int_{-\infty}^{t} i(\tau)\,d\tau.$$

Systems: Causal and Noncausal

A system is **causal** if the output at time t_0 depends only on the input for $t \le t_0$, i.e., the system cannot anticipate the input.

- Causality

A CT system $x(t) \to y(t)$ is causal if

When $x_1(t) \to y_1(t)$ $x_2(t) \to y_2(t)$

and $x_1(t) = x_2(t)$ $for\ all\ t \le t_0$

Then $y_1(t) = y_2(t)$ $for\ all\ t \le t_0$

Systems: Causal and Noncausal

Ex.#1 $y(t) = x(t-1)$

Ex.#2 $y(t) = x(t+1)$

Ex.#3 $y[n] = x[-n]$

Ex.#4 $y[n] = \left(\frac{1}{2}\right)^{n+1} x^3[n-1]$

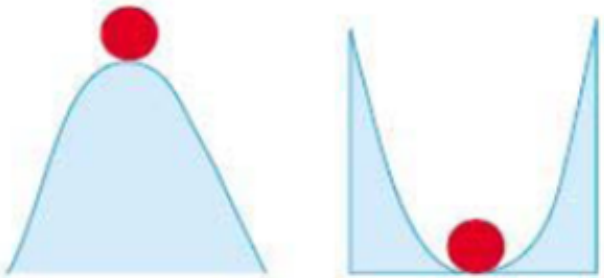
Ex.#5 $y(t) = x(t)\cos(t+1)$

Systems: Causal and Noncausal

- All real-time physical systems are causal, because time only moves forward. Effect occurs after cause. (Imagine if you own a noncausal system whose output depends on tomorrow's stock price.)
- Causality does not apply to spatially varying signals. (We can move both left and right, up and down.)
- Causality does not apply to systems processing recorded signals, e.g. taped sports games vs. live broadcast.

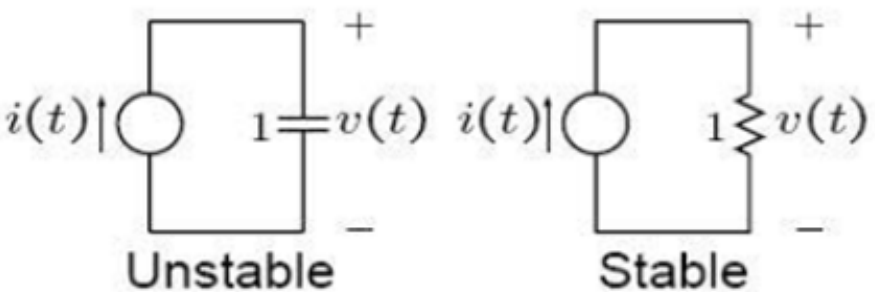
Systems: Stable and Non-stable

Stability can be defined in a variety of ways.
Definition 1: a stable system is one for which an incremental input leads to an incremental output.



An incremental force leads to only an incremental displacement in the stable system but not in the unstable system.

Definition 2: A system is BIBO stable if every bounded input leads to a bounded output. We will use this definition.



For the resistor, if $i(t)$ is bounded then so is $v(t)$, but for the capacitance this is not true. Consider $i(t) = u(t)$ then $v(t) = tu(t)$ which is unbounded.

Time-invariant Systems

Informally, a system is time-invariant (TI) if its behavior does not depend on the choice of $t = 0$. Then two identical experiments will yield the same results, regardless the starting time.

- Mathematically (in DT): A system is TI if for *any* input $x[n]$ and *any* time shift n_0,

$$\text{If} \quad x[n] \rightarrow y[n]$$
$$\text{then} \quad x[n - n_0] \rightarrow y[n - n_0] \, .$$

- Similarly for CT time-invariant system,

$$\text{If} \quad x(t) \rightarrow y(t)$$
$$\text{then} \quad x(t - t_0) \rightarrow y(t - t_0) \, .$$

Time-invariant Systems

Ex.#1 $y(t) = x^2(t+1)$

Ex.#2 $y[n] = \left(\frac{1}{2}\right)^{n+1} x^3[n-1]$

Ex.#3 $y(t) = \sin[x(t)]$

Ex.#4 $y[n] = nx[n]$

Ex.#5 $y(t) = x(2t)$

Time-invariant Systems

- If the input to a TI System is periodic, then the output is periodic with the same period.

"Proof": Suppose $x(t + T) = x(t)$

and $x(t) \rightarrow y(t)$

Then by TI

$$x(t + T) \rightarrow y(t + T).$$

These are the same input!

So these must be the same output, i.e., $y(t) = y(t + T)$

Linear Systems

A (CT) system is linear if it has the superposition property:

If $x_1(t) \rightarrow y_1(t)$ and $x_2(t) \rightarrow y_2(t)$

then $ax_1(t) + bx_2(t) \rightarrow ay_1(t) + by_2(t)$

$y[n] = x^2[n]$ Nonlinear, TI, Causal

$y(t) = x(2t)$ Linear, not TI, Noncausal

Can you find systems with other combinations ?

- e.g. Linear, TI, Noncausal
  Linear, not TI, Causal

Linear Systems

- Superposition

If $x_k[n] \rightarrow y_k[n]$

Then $\sum_k a_k x_k[n] \rightarrow \sum_k a_k y_k[n]$

- For linear systems, zero input $\rightarrow$ zero output

"Proof" $0 = 0 \cdot x[n] \rightarrow 0 \cdot y[n] = 0$

Outline

1. Course Introduction
2. Introduction to Signals and Systems
3. Introduction to Signals
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5. Assignments

Assignments

- Reading Assignment: Chapter 1.0-1.6